

Jakub Sokal

jakubsokal13@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Profile Summary

Software engineer with 12 months of commercial experience at AMCS Group, working on live payment systems, diagnosing a critical double-charge defect in production and building end-to-end Stripe and SEPA integrations. Named student researcher on the EU-funded Prompt Me project at Lero, University of Limerick, designing and building a multi-agent LLM orchestration framework for AI-driven Requirements Engineering (RE). Hands-on experience building and optimising backend systems under real production constraints with focus on large-scale infrastructure, distributed systems, and fintech. Graduating August 2026.

Education

University of Limerick

Limerick, Co. Limerick

Bachelor of Science in Computer Science | Expected August 2026 | First Class Honours (QCA: 3.44/4.0)

Course Work: Database Systems, Advanced Programming Concepts and Practices, Data Structures & Algorithms, Software Architecture, Applied System Design, Operating Systems, Software Testing and Inspection, Event Driven Programming.

Professional Experience

AMCS Group

Limerick, Co. Limerick

Associate Software Developer

June 2025 – August 2025

- Designed and delivered production Stripe Card and SEPA payment integrations from technical design through to deployment.
- Diagnosed and fixed a critical double-charge defect in the live payment flow by tracking down the root cause, patching it and shipping to production.
- Optimised a transaction endpoint handling 5,000+ records, reducing response time by 40% from approximately 10 minutes to under 6 minutes.
- Identified and documented the root cause of a serialization bottleneck in the RESTful API layer, producing an analysis that pinpointed the exact failure point and directly enabled the fix.
- Built Azure Data Factory pipelines for ETL processes and managed CI/CD deployments across dev, test, and production Azure environments.

AMCS Group

Limerick, Co. Limerick

Intern Software Developer

May 2024 – January 2025

- Developed bug fixes and feature enhancements for Eremos, a legacy WPF desktop application, working within an existing C#/.NET codebase.
- Built out unit test coverage across core payment and API components in a large-scale C#/.NET and Angular codebase, writing 500+ tests and consistently maintaining the team's 80% per-PR coverage requirement.
- Worked on ACI Cards payment integration throughout a live refactoring cycle, shipped bug fixes, features, and tests while the code base was actively changing.
- Traced a data integrity bug in the Payment Express feature back to an upstream service misappending results, identifying the root cause of missing column data in the client-facing UI and documenting findings for resolution.
- Extended and maintained 30+ RESTful APIs across live payment and data flows, shipping to production across multiple sprint cycles.
- Worked in pair programming sessions with senior developers, collaborating on feature development and code quality improvements.

Project Experience

Final Year Project (SimOrch)

University of Limerick

Orchestration Framework for Multi-Agent RE Simulation | [GitHub](#) | [Link](#) | Python | React | Typescript

September 2025 – April 2026

- Named student researcher on the EU-funded Prompt Me project at Lero, University of Limerick, supervised by Dr. Jacek Dabrowski, investigating the feasibility of AI-driven Requirements Engineering simulation.
- Designed and built a multi-agent orchestration framework in Python coordinating GPT-4-powered Requirements Engineer, Stakeholder, and Analyst Agents with shared memory, persona-based configuration, and structured logging to produce reproducible and traceable RE simulation outputs.
- Evaluated the framework across seven dimensions, achieving 100% feasibility and 100% traceability across all ten simulation runs, with structural reproducibility of 100% and ground truth requirement coverage of 83.3%.
- Identified a limitation in the current simulation: LLM-based RE agents failed to elicit non-functional requirements under the tested persona configuration, highlighting an area for further investigation in dialogue-driven elicitation.
- Research findings contributed to an in-progress international collaboration spanning University of Limerick, Trinity College Dublin, and The Open University.
- Built a FastAPI backend with WebSocket support and polling endpoints to deliver real-time simulation updates to the frontend.

Software Architecture Project (Campus2Company)

University of Limerick

Microservices Platform | [GitHub](#) | Java | Spring Boot | React | Kafka| Docker

February 2026 – April 2026

- Led development across the team, owning code reviews and driving architecture decisions from initial design through to completion.
- Designed and built a Java Spring Boot microservices architecture with an API gateway to decouple services and manage inter-service communication.
- Implemented a full JWT authentication system including access tokens, refresh tokens, and long-lived tokens for persistent session management across services.
- Integrated Kafka for asynchronous event-driven communication between microservices, decoupling producers and consumers across the platform.
- Containerised core services using Docker to ensure consistent deployments across environments.
- Built a responsive React frontend supporting interactive workflows across multiple user roles such as students, universities, and businesses.

Skills

Languages: Java, Python, C#, JavaScript, TypeScript, C++, Go.

Frontend: Next.js, React, Angular, WPF, HTML, CSS.

Backend: RESTful APIs, FastAPI, Spring Boot, .NET Framework, Entity Framework, Kafka, Redis, SQL, PostgreSQL, Supabase, JWT Authentication.

Testing: Postman, NUnit, JUnit, Jasmine, pytest, TDD.

AI / LLM: LangChain, OpenAI API, PyTorch, Numpy, Pandas, Multi-Agent Orchestration.

Tools: Plotly, Jupyter Notebook.

Infrastructure & Tools: CI/CD, Docker, Azure, Git, Azure Data Factory.

Soft Skills: Problem Solving, Critical Thinking, Communication, Teamwork, Agile Methodologies, Attention to Detail, Adaptability.

References

Available upon request.